

Causal attention

	The	cat	sat	on	the
The	1	0	0	0	0
cat	1	1	0	0	0
sat	1	1	1	0	0
on	1	1	1	1	0
the	1	1	1	1	1

Each query attends to **every** previous token and itself — lower-triangular mask.

Attention + sink token

	The	cat	sat	on	the
The	1	0	0	0	0
cat	1	1	0	0	0
sat	1	1	1	0	0
on	1	1	1	1	0
the	1	1	1	1	1

↑
sink token

Every query also attends the first (sink) token — the softmax dumps spare mass there.

$\text{softmax}([q \cdot k_0 \cdots q \cdot k_i, s])$ — s is a learned per-head sink logit (no value vector)

attended (1)

 sink column (always on)

 masked (0)